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Research paper

MAM-PhyGNN: A Physics-Hardcoded Graph Neural Network for fast and accurate thermal simulation in metal additive manufacturing

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ABSTRACT

Accurate and efficient prediction of transient thermal fields in metal additive manufacturing (MAM) is critical for process control, but a persistent trade-off exists between fidelity and speed. High-fidelity solvers require hours of computation for a single simulation, while existing surrogates like Physics-Informed Neural Networks (PINNs) often fail in long-term predictions due to their topology-agnostic nature and soft physical constraints. To overcome these limitations, this paper introduces the Metal Additive Manufacturing Physics-Hardcoded Graph Neural Network (MAM-PhyGNN), a topology-aware Graph Neural Network (GNN) that architecturally embeds physical laws directly onto the unstructured simulation mesh. While the framework is designed for general MAM processes, its performance is validated on a challenging Laser Powder Bed Fusion (LPBF) case study. The framework formulates the heat transfer problem on an unstructured mesh as a dynamical system on a graph. Its core innovation is a dual-branch operator architecture: a deterministic, non-learnable Laplace Block hardcodes the fundamental physics of heat diffusion, providing a robust stability anchor, while a data-driven GNN learns a data-driven correction for complex nonlinear dynamics arising from temperature-dependent material properties and boundary conditions. In the LPBF case study, MAM-PhyGNN demonstrated exceptional performance. It remained stable through a 1000-step autoregressive long-term prediction on unseen data, maintaining a Pearson correlation of 0.98 with the ground truth. Crucially, MAM-PhyGNN achieved this with a computational speedup of approximately 850x over the high-fidelity solver, reducing simulation time from over 50 min to just 3.6 s. This work demonstrates that by architecturally embedding fundamental physics within a topology-aware GNN, it is possible to create surrogate models that are simultaneously fast, accurate, and stable for long-duration MAM simulations, thereby opening a viable pathway towards real-time digital twins for process optimization.

1. Introduction

Additive Manufacturing (AM) is revolutionizing the production of high-performance components across aerospace, medical, and energy sectors, enabling the creation of parts with unprecedented geometric complexity and tailored material properties [1,2]. Among its various modalities, MAM—particularly processes like LPBF—is pivotal for creating such high-performance metallic components. It is well-established that the final quality of an MAM part—including its microstructure, residual stresses, and defect formation—is critically dictated by its transient thermal history during the fabrication process [3].

Recognizing this thermal dependence, researchers have increasingly deployed in-situ monitoring systems in MAM processes to track real-time thermal signatures and assess build quality based on thermal history [4,5]. Current in-situ monitoring systems primarily focus on detecting existing thermal anomalies through sensors such as infrared cameras. While these systems can identify overheating or insufficient fusion after they occur, they lack the capability to predict future thermal states. Crucially, effective process control requires not just defect detection [6], but defect prevention. This necessitates predicting the temperature field evolution before anomalies develop, enabling proactive parameter adjustments that maintain optimal thermal conditions throughout the build process.

To enable such predictive capabilities, high-fidelity numerical simulations like Finite Element Analysis (FEA) offer a theoretically sound approach for forecasting thermal evolution. However, these methods face a fundamental limitation: computational speed. A single FEA simulation typically requires hours or days to complete [7], making it incompatible with real-time control requirements where predictions

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must be generated within milliseconds to match manufacturing cycle times. This creates an urgent need for surrogate thermal models that satisfy three critical conditions simultaneously: (1) physical accuracy sufficient to capture key thermal phenomena, (2) computational speed compatible with real-time decision-making (milliseconds per prediction), and (3) long-term stability for sustained operation during extended builds.

Recent advances confirm that when these requirements are met, real-time thermal control becomes feasible. For instance, closed-loop laser power control systems have demonstrated that millisecond-level thermal predictions enable proactive parameter adjustments that prevent defect formation before it occurs [8,9]. This establishes a clear principle: computational speed is the primary enabler of defect prevention in MAM processes. The urgent challenge, therefore, is developing thermal prediction frameworks that maintain physical fidelity while achieving the millisecond response times required for real-time process control—a challenge that existing surrogate modeling approaches have yet to address comprehensively.

Machine learning has emerged as a promising avenue for developing such surrogates, yet mainstream approaches exhibit fundamental limitations when applied to AM physics [10–12]. For example, many efforts follow a traditional surrogate modeling paradigm: using high-fidelity simulations to generate a dataset and then training a data-driven model, such as an Artificial Neural Network (ANN), to predict a specific quantity of interest. Farias et al. successfully employed this strategy to create an ANN that predicts interpass temperature from process parameters, enabling the creation of a process map without repeated, time-consuming FEM simulations [13]. Other works use more complex architectures; for instance, Nalajam et al. proposed a hybrid model combining a Convolutional Neural Network (CNN) for spatial feature extraction and a Long Short-Term Memory (LSTM) network for temporal dependency modeling [14,15]. While such methods can improve accuracy for specific tasks like melt pool temperature prediction, they have two fundamental constraints. First, their reliance on structured, grid-like data creates a fundamental trade-off between accuracy and computational efficiency for thermal prediction. To resolve the extreme thermal gradients around the melt pool, a CNN-based model requires a uniformly fine grid across the entire domain, leading to prohibitive computational costs. This rigid structure precludes the use of adaptive local refinement, a key strategy for efficiently modeling localized phenomena without incurring excessive computational overhead. Second, as purely data-driven models, they lack mechanisms to enforce physical laws, which can lead to unphysical predictions and instability over long time horizons.

To address the lack of physical constraints in purely data-driven models, Physics-Informed Neural Networks (PINNs) were developed [16–18]. PINN embeds physical laws by minimizing a PDE residual within its loss function, a technique often termed a “soft constraint” [19]. This paradigm has been successfully applied to MAM, with notable contributions from Zhu et al. who first used PINNs for MAM thermal prediction [12], and Liao et al. who developed a hybrid physics-data driven PINN for parameter identification [20]. Further research has focused on adapting PINNs to the specific challenges of MAM, such as the intrinsic discontinuities of the layer-by-layer process. Peng and Panesar, for instance, developed a multi-layer PINN framework for DED that uses specialized strategies, including pointwise weighting for initial conditions and smooth activation functions for solid-void transitions, to manage these discontinuities [21]. Other works have focused on reducing the reliance on labeled data while maintaining high accuracy and efficiency [22].

Despite these successes, the conventional PINN formulation suffers from two limitations that hinder its use for robust, dynamic MAM thermal simulation. First, the reliance on a “soft constraint” is fundamentally an optimization challenge [23]. Minimizing the PDE residual provides no guarantee of converging to the true physical solution, as the optimization process can be trapped in local minima or satisfy the loss with non-physical, trivial solutions. Second, and more critically, the conventional MLP-based PINN architecture is topology-agnostic. It processes spatiotemporal coordinates as a collection of independent points, lacking any intrinsic understanding of the spatial relationships that govern heat transfer [24]. This is particularly problematic for MAM thermal simulation, where the physics is dominated by highly localized heat conduction radiating from a moving laser source [25]. An MLP cannot inherently distinguish between the critical heat exchange occurring between adjacent nodes in the melt pool and the less significant interactions in cooler, distant regions. It is forced to learn this fundamental concept of local connectivity from data alone, an inefficient process that struggles to capture the underlying structure of the PDE operator. As our results will show, this architectural limitation severely hinders its ability to make stable, long-term predictions, as small errors can accumulate without the stabilizing influence of a well-defined local structure.

An alternative strategy to address this topological blindness involves embedding spatial awareness not in the model architecture, but through intelligent feature engineering. Roy and Wodo demonstrated this by translating G-code directly into a set of physics-aware features, such as local distances from heat sources and sinks [13]. By using an analytically-derived “Heat Influence Zone” (HIZ) to select the most relevant features, they trained a standard ANN to predict the full thermal history. This approach cleverly circumvents the need for explicit geometry and acknowledges the importance of local interactions. However, because the physical intuition is encoded in the input features rather than the model itself, the network remains a purely data-driven learner, which may still limit its generalization and long-term stability.

To address these limitations more fundamentally, research has turned to GNN as a more suitable architecture. Because GNNs operate on unstructured meshes, they are inherently topology-aware, providing a natural framework for modeling the local interactions central to physical processes like heat diffusion. For instance, recent work by Mozaffar et al. has demonstrated that a GNN-based approach can successfully model the thermal history in AM processes across varying geometries, leveraging the graph structure to capture spatial topology and improve generalization [26]. However, these pioneering GNN models are often purely data-driven. Like the CNN-LSTM models mentioned earlier, they lack explicit physical constraints, which can limit their own stability and physical consistency, particularly in long-term forecasting. Therefore, a significant gap remains for a surrogate modeling paradigm that is both inherently topology-aware and enforces physical principles with greater rigor than either a soft loss-term constraint or a purely data-driven approach.

This work bridges that gap by proposing the MAM-PhyGNN, a framework that is both topology-aware and physically-rigorous. Our core innovation is to move beyond soft constraints and architecturally embed the governing physics directly into the network’s structure. This is achieved through a novel dual-branch architecture that decomposes the physical operator into a primary physics-driven component and a data-driven corrective component. A physics-hardcoded Laplace Block, based on the analytical discrete Laplace–Beltrami operator, computes the dominant linear heat diffusion in a deterministic and non-learnable manner, providing a robust physical backbone for the model. This yields a fast and stable, but imperfect, first approximation of the thermal evolution. Concurrently, a data-driven GNN Block learns to predict the “residual dynamics”—a corrective term that accounts for what the simplified physics model misses. This includes complex nonlinear effects, such as temperature-dependent material properties, as well as compensating for numerical discretization errors. The framework is explicitly tailored to handle the non-stationary moving heat source characteristic of MAM.

The main contributions of this work are three-fold:

- A novel framework that formulates the transient heat transfer PDE as a dynamical system on a graph. This reconceptualization,

which applies the principles of the Method of Lines (MOL, a numerical technique that discretizes space to convert a PDE into a system of ODEs), allows a GNN to learn the temporal evolution of the system, directly addressing the topological and physical limitations of prior surrogate models.
- A dual-branch, physics-hardcoded GNN architecture is introduced, which strategically separates linear, deterministic physics from learnable nonlinear dynamics to ensure both physical consistency and predictive accuracy.
- The state-of-the-art stability and accuracy of the MAM-PhyGNN model is demonstrated through rigorous long-term autoregressive predictions. In long-term autoregressive prediction, MAM-PhyGNN remains stable for over 1000 time steps on unseen data, reduces prediction error by an order of magnitude compared to conventional PINNs, and achieves a computational speedup of approximately 850× over the high-fidelity solver.

The remainder of this paper is structured as follows. Section 2 details the governing physics of the AM thermal problem and the core methodology of the MAM-PhyGNN framework, including the graph construction, the dual-branch message-passing architecture, and the temporal integration scheme. Section 3 presents a comprehensive case study on LPBF, where the model's performance is evaluated through generalization tests, long-term prediction stability analysis, and a series of ablation and comparative experiments. Section 4 then discusses the broader implications of these findings, analyzing the mechanisms behind the model's superior performance. Finally, Section 5 summarizes the study and outlines the directions for future work. The code framework will be publicly available at https://github.com/zhanghuaqing123/Paper-MAM-PhyGNN upon formal publication of the paper.

2. Methodology: The MAM-PhyGNN framework

While high-fidelity FEM solvers are accurate, their computational cost is prohibitive for real-time applications, and existing surrogate models suffer from critical limitations in physical consistency and topological awareness. Our framework, MAM-PhyGNN is designed to be both inherently topology-aware and rigorously physics-enforced. By operating on a graph representation of the unstructured mesh, it explicitly models the local connectivity between interacting points in the domain, which is fundamental to the physics of heat conduction. More critically, our core innovation is to move beyond soft constraints and architecturally "hardcode" the governing physics directly into the network. This is achieved by decomposing the physical operator into a deterministic, fixed numerical component for linear diffusion and a learnable GNN for nonlinear residuals, ensuring both physical consistency and predictive accuracy.

This synergistic integration of numerical methods and machine learning transforms the transient heat transfer simulation into a learning problem on a graph-based dynamical system. As illustrated in Fig. 1, the methodology comprises five key stages: (1) defining the governing physical model, (2) transforming the model into a dimensionless form, (3) representing the spatial domain as a computational graph, (4) approximating the system's physical operator with our novel dual-branch GNN, and (5) advancing the system in time with a high-order numerical integrator. The following sections will detail each of these stages.

2.1. The governing physical model

The physical foundation of our model is the transient heat transfer process common to MAM techniques driven by a moving laser, such as Directed Energy Deposition (DED) and LPBF. This process is described by a nonlinear partial differential equation (PDE) for heat conduction, coupled with boundary conditions that model heat exchange with the environment. This physical system, shown in Fig. 1(a), serves as the starting point for our framework.

The transient heat conduction in MAM is governed by the following nonlinear PDE:

$$\rho(T)c_p(T)\frac{\partial T(\mathbf{x},t)}{\partial t} = \nabla\cdot(k(T)\nabla T(\mathbf{x},t)) + Q_{src}(\mathbf{x},t) \tag{1}$$

where $T(\mathbf{x},t)$ is the temperature field, while $\rho(T)$, $c_p(T)$, and $k(T)$ are the temperature-dependent density, specific heat, and thermal conductivity, respectively. This formulation captures the full nonlinearity of the problem. To illuminate its structure, we can contrast it with a simplified, linearized version. If the material properties are assumed constant, the equation reduces to:

$$\frac{\partial T}{\partial t} = \alpha\Delta T + \frac{Q_{src}}{\rho c_p} \tag{2}$$

where $\alpha = \frac{k}{\rho c_p}$ is the constant thermal diffusivity. In this case, the heat diffusion term is directly proportional to the Laplacian operator ΔT. $Q_{src}(\mathbf{x},t)$ models a moving Gaussian laser heat flux, the detailed formulation of which is provided in Section 2.4.4.

At the domain boundary, we impose a Robin condition to account for combined convection and radiation with the surroundings at an ambient temperature, T_{amb}:

$$-k\nabla T\cdot\mathbf{n} = h_c(T-T_{amb}) + \varepsilon\sigma(T^4 - T_{amb}^4) \tag{3}$$

where h_c is the convective coefficient, ε is the emissivity, and σ is the Stefan–Boltzmann constant. $\mathbf{n}$ is the outward-facing normal vector.

2.2. Dimensionless formulation for generality and stability

To create a more general and numerically stable problem, we nondimensionalize the governing equations, a process summarized in Fig. 1(b). This step improves model convergence and enhances its generalization capability across different scales and materials.

The process begins by defining a set of characteristic scales based on the problem's physics [27]:

- **Characteristic Length**, L_{ref}: A representative dimension of the system, such as the melt pool width.
- **Characteristic Temperature**, T_{ref}: The material's liquidus temperature, a key physical threshold.
- **Reference Thermal Properties**: Material properties evaluated at T_{ref}, such as k_{ref}, $c_{p,ref}$, and ρ_{ref}.
- **Characteristic Time**, t_{ref}: The derived time scale for heat diffusion, defined as $t_{ref} = \frac{L_{ref}^2}{\alpha_{ref}}$.

Using these scales, all physical variables are transformed into their dimensionless counterparts (denoted by a superscript $*$), such as $\mathbf{x}^* = \frac{\mathbf{x}}{L_{ref}}$, $T^* = \frac{T}{T_{ref}}$, $t^* = \frac{t}{t_{ref}}$, $dt^* = \frac{dt}{t_{ref}}$. Substituting these into the governing equations yields their dimensionless forms. For instance, the simplified heat equation becomes:

$$\frac{\partial T^*}{\partial t^*} = \Delta^* T^* + Q_{src}^* \tag{4}$$

where Δ^* is the Laplacian operator in dimensionless coordinates and $Q_{src}^* = \frac{L_{ref}^2}{k_{ref}T_{ref}}\cdot Q_{src}$ is the dimensionless heat source.

The Robin boundary condition transforms into:

$$-k^*\nabla^* T^*\cdot\mathbf{n}^* = \mathrm{Bi}(T^* - T_\infty^*) + \mathrm{Sk}(T^{*4} - T_\infty^{*4}) \tag{5}$$

This formulation elegantly reveals the key dimensionless groups governing the system's boundary behavior:

- **The Biot number**, $\mathrm{Bi} = \frac{h_c L_{ref}}{k_{ref}}$, which compares the rate of heat transfer by convection to that by conduction.

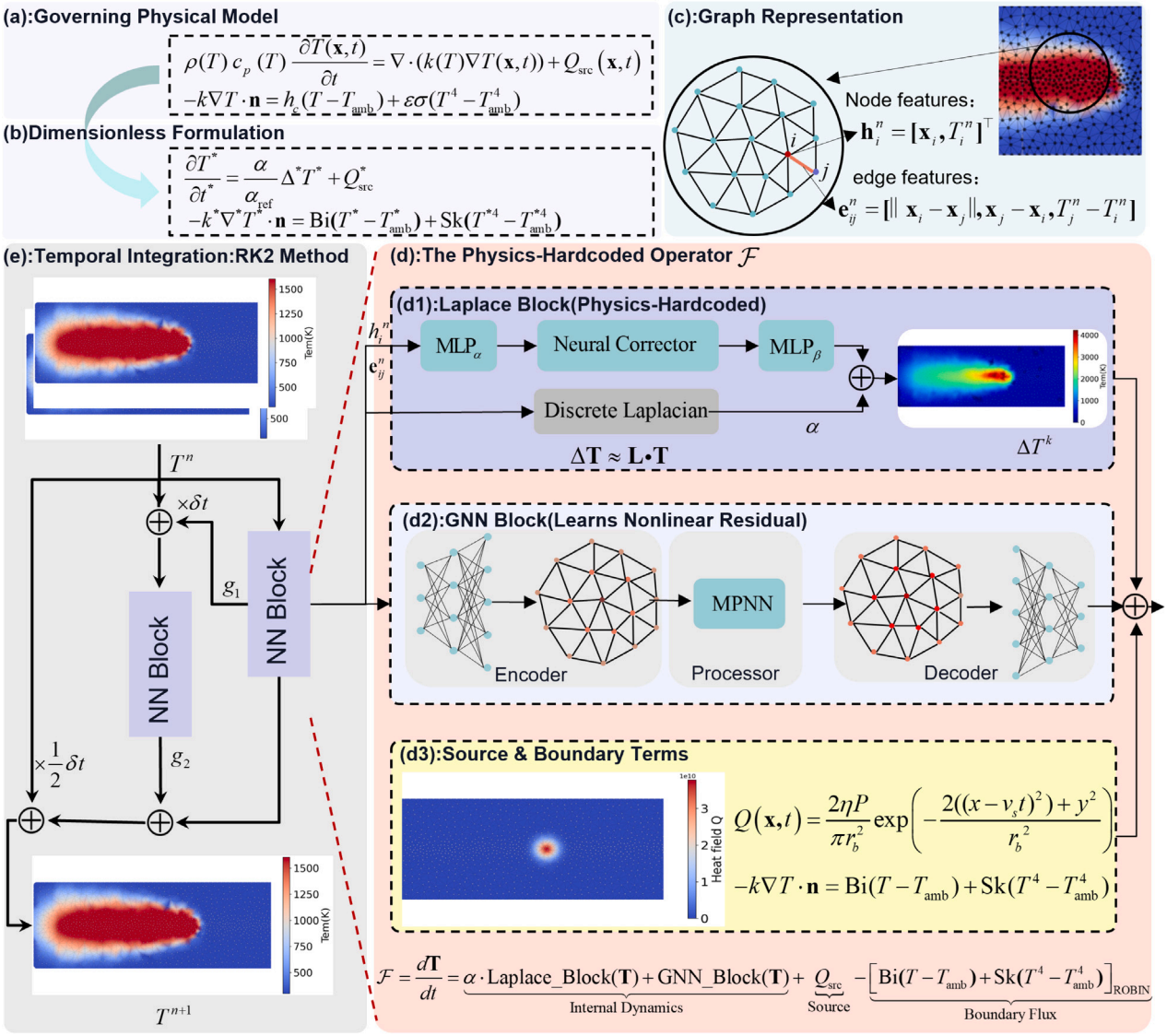


Fig. 1. The MAM-PhyGNN simulation framework. The methodology transforms the continuous physical problem into a discrete, learnable simulation loop. (a) The process begins with the governing physical model, including the dimensional heat equation and Robin boundary condition. (b) This model is then transformed into a dimensionless form to improve stability and generality. (c) The spatial domain is discretized into a computational graph, with node and edge features encoding the system's physical state. (d) The core of the framework is a physics-hardcoded operator, which approximates the system's time derivative (dT/dt). It decomposes the dynamics into: (d1) a hardcoded Laplace Block for linear diffusion, (d2) a learnable GNN Block for nonlinear residuals, and (d3) explicit external heat source terms and boundary terms. (e) Finally, a second-order Runge–Kutta (RK2) integrator uses the operator to advance the temperature field in time, with an autoregressive loop enabling long-horizon transient simulation.

- **The radiation parameter (or Stark number)**, $\mathrm{Sk} = \frac{\varepsilon \sigma T_{ref}^3 L_{ref}}{k_{ref}}$, which compares the rate of heat transfer by radiation to that by conduction.

Having established the dimensionless formulation, for the sake of notational clarity, we will henceforth drop the $*$ superscript. Unless explicitly stated otherwise, all physical quantities and operators discussed in the following sections are to be considered in their dimensionless form.

2.3. Discretization: From continuous PDE to a learnable dynamical system

With the dimensionless physical model established, we now discretize it to create a computational framework suitable for a GNN.

2.3.1. Spatial discretization via graph construction

To solve the governing PDE numerically, we must first discretize the continuous physical domain. We achieve this by representing the domain as a computational graph, a structure perfectly suited for a GNN. This process, illustrated in Fig. 1(c), involves two main steps: creating a 2D mesh of the temperature field and then mapping this mesh into a physics-aware graph.

First, we model the problem by considering a 2D cross-section of the MAM build, such as a thin layer within a LPBF process. This geometry is then discretized into an unstructured triangular mesh, which serves as the computational grid for the 2D temperature field. This mesh acts as the scaffold for our graph. We then directly map this mesh into a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes and edges are designed to encode AM-specific physical information, transforming the raw geometric data into a structured format amenable to neural learning. This mapping process is visually detailed in Fig. 1(c).

- **Nodes** ($\mathcal{V}$): Each node $i \in \mathcal{V}$ corresponds to a vertex in the discretized mesh. At each time step t_n, the node carries a node feature vector $\mathbf{h}_i^n$ that encapsulates all key physical quantities required to drive the thermal simulation:

$$\mathbf{h}_i^n = [\mathbf{x}_i, T_i^n] \tag{6}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ are the spatial coordinates of node i (where $d = 2$ or 3), defining the system's geometry. T_i^n is the temperature of the node at the current time step t_n, serving as the core state variable of the model.

- **Edges** ($\mathcal{E}$): Each edge $(i, j) \in \mathcal{E}$ represents the connectivity between two vertices i and j in the mesh. This ensures that the pathways for heat flow in the graph are identical to the physical connectivity of the domain. The edge features $\mathbf{e}_{ij}^n$ encode the geometric and physical relationship between the connected nodes:

$$\mathbf{e}_{ij}^n = [\|\mathbf{x}_i - \mathbf{x}_j\|, \mathbf{x}_j - \mathbf{x}_i, T_j^n - T_i^n] \tag{7}$$

  (1) $\|\mathbf{x}_i - \mathbf{x}_j\|$: The Euclidean distance between the nodes, which is fundamental for calculating heat flux.
  (2) $\mathbf{x}_j - \mathbf{x}_i$: The directional vector from node i to node j, providing geometric information for heat propagation.
  (3) $T_j^n - T_i^n$: The temperature difference between the nodes in time step t_n, which directly represents the local temperature gradient that drives heat conduction.

Through this structured approach, we transform the raw geometry into a physics-rich graph. This graph not only preserves the topology of the 2D temperature field but also provides a complete and physically meaningful set of inputs for the subsequent message-passing framework, enabling it to learn the underlying thermal dynamics.

2.3.2. Spatio-temporal semi-discretization via Method of Lines (MOL)

Next, we employ the Method of Lines (MOL) to transform the PDE into a system of ordinary differential equations (ODEs). By approximating the spatial derivatives on the graph nodes, the original PDE is converted into a large system of coupled ODEs, one for each node i:

$$\frac{dT_i}{dt} = \Delta T_i + Q_{src,i} = \mathcal{F}(T_i) \tag{8}$$

Here, T_i is the temperature of node i, and $\mathcal{F}$ is a nonlinear physical operator that represents the rate of change of temperature, encapsulating all physical effects such as heat diffusion, heat sources, and boundary conditions. With this transformation, the simulation task becomes equivalent to learning the operator $\mathcal{F}$ that governs the system's evolution. Once this operator is learned, any standard numerical integration scheme can be used to autoregressively predict the temperature field over time. Consequently, our primary task is to design a neural network that accurately and efficiently approximates this operator $\mathcal{F}$.

2.4. Constructing the physics-hardcoded operator

Having established the ODE system through the Method of Lines (Section 2.3.2), we now introduce the core innovation of this work: a physics-hardcoded operator $\mathcal{F}$ that computes the instantaneous rate of temperature change across the domain. Rather than treating $\mathcal{F}$ as a monolithic black-box function, our approach decomposes it into physically interpretable components that directly correspond to fundamental heat transfer mechanisms. As shown in Fig. 1(d), the complete operator integrates three complementary elements: (1) a deterministic Laplace Block that exactly encodes linear heat diffusion physics, (2) a learnable GNN Block that captures complex nonlinear residuals, (3) explicit terms for external heat sources and boundary condition contributions. This decomposition strategy ensures that well-understood physics remain deterministic numerical approximations, while data-driven components focus exclusively on modeling phenomena that lack closed-form solutions.

The following subsections develop this architecture systematically. We begin with the mathematical foundation of heat diffusion—the discrete Laplace–Beltrami operator—before detailing the hybrid design of the Laplace Block. Next, we present the GNN Block's architecture for learning nonlinear dynamics, followed by explicit formulations for external sources and boundary conditions. Finally, we unify these components into the complete physics-hardcoded operator, which is then advanced in time using a high-order numerical integrator. This structured approach provides both physical interpretability and computational efficiency while maintaining strict adherence to thermodynamic principles.

2.4.1. The discrete Laplace–Beltrami operator: A physical cornerstone for diffusion

To approximate $\mathcal{F}$ effectively, we first isolate its most dominant and well-understood component: the heat diffusion term ΔT. It mathematically describes the difference between a point and its surroundings, representing the fundamental driving force of heat flow from hotter to colder regions. To accurately represent this crucial physical process on our unstructured mesh, we employ the Laplace–Beltrami operator, which generalizes the Laplacian to our mesh geometry. On a 2D triangular mesh, a widely adopted method that preserves key physical properties is the cotangent formula:

$$\Delta T_i \approx \sum_{j \in \mathcal{N}(i)} w_{ij}(T_j - T_i) \tag{9}$$

where

$$w_{ij} = \frac{1}{2A_i}(\cot \alpha_{ij} + \cot \beta_{ij}) \tag{10}$$

Each component of this formula has a clear physical and geometric meaning, as illustrated in Fig. 1(c):

- $T_j - T_i$ is the temperature difference between node i and its neighbor j, which is the driving force for heat flux.
- $w_{ij} = \frac{1}{2A_i}(\cot \alpha_{ij} + \cot \beta_{ij})$ is the edge weight of edge (i, j). It is determined by the cotangents of the angles α_{ij} and β_{ij}, where α_{ij} and β_{ij} are the two angles opposite to the shared edge (i, j) (Fig. 2(a)). It acts like a "thermal conductance", reflecting the heat transfer efficiency of the geometric connection between nodes.
- A_i is the dual area associated with node i, physically representing its effective "thermal mass region". It is calculated as the Voronoi area, defined by the polygon connecting the circumcenters of the triangles surrounding node i (Fig. 2(b)–(d)).

The power of this formulation is that it describes a linear operation on the temperature field, which can be elegantly expressed as a global matrix–vector product. To do this, we assemble a sparse $N \times N$ mesh Laplacian matrix $\mathbf{L}$, where N is the total number of nodes. The elements of $\mathbf{L}$ are defined directly from the cotangent formula's components:

Off-diagonal elements ($i \neq j$): For each pair of connected nodes i and j, the element L_{ij} captures their interaction. It is defined as the edge weight w_{ij} scaled by the source node's area A_i. If two nodes are not connected, their corresponding element is zero.

$$L_{ij} = \begin{cases} \frac{w_{ij}}{A_i} & \text{if } j \in \mathcal{N}(i) \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

Diagonal elements ($i = j$): Each diagonal element L_{ii} represents the central node's contribution and is defined as the negative sum of all other elements in its row. This ensures that for a uniform temperature field (where all $T_j = T_i$), the Laplacian is zero, correctly reflecting a state of equilibrium.

$$L_{ii} = -\sum_{j \neq i} L_{ij} = -\sum_{j \in \mathcal{N}(i)} \frac{w_{ij}}{A_i} \tag{12}$$

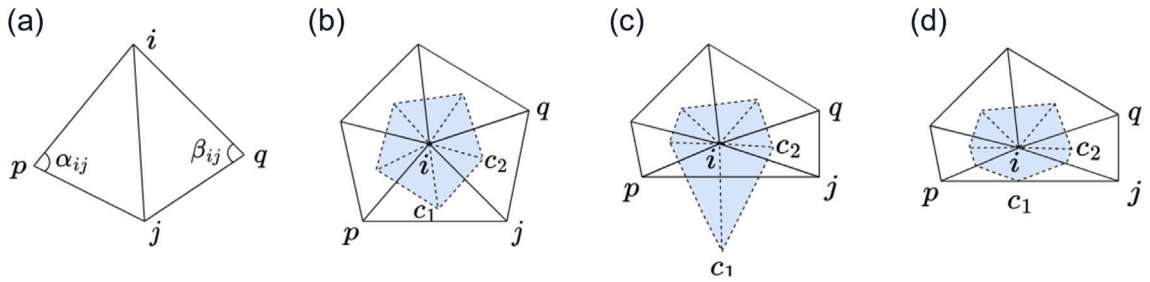


Fig. 2. Geometric interpretation of the components for the discrete Laplace–Beltrami operator on a triangular mesh. (a) The edge weight w_{ij} is calculated from the cotangents of the angles α_{ij} and β_{ij}, which are opposite to the shared edge (i, j). (b) The dual area A_i (shaded region) associated with node i is its Voronoi cell. (c–d) The construction of this dual area is shown by connecting the circumcenters (c_1, c_2) of the adjacent triangles.

With the matrix **L** constructed, applying the diffusion operator to the entire temperature vector **T** (a vector of size N) simplifies to a single, highly efficient sparse matrix–vector product:

$$\Delta \mathbf{T} \approx \mathbf{LT} \tag{13}$$

By doing so, we pre-encode the core diffusion physics into a fixed, non-learnable linear operator **L**. This provides our model with an extremely robust and physically consistent baseline for the linear, isotropic heat diffusion process, forming the cornerstone of the Laplace Block detailed next. While this deterministically defined operator captures the dominant parabolic behavior, it cannot model the full spectrum of physics defined in Section 2.1—specifically, the nonlinearities and the moving source. Therefore, we complement it with a physics-driven GNN, tasked with learning the complex residual dynamics that the Laplacian alone cannot represent. This decomposition establishes the blueprint for our solver: a dual-branch architecture where a fixed Laplace Block computes the linear diffusion baseline, while a learnable GNN Block approximates the nonlinear residual. This design ensures that fundamental physics are hardcoded, while data-driven components capture intractable phenomena.

2.4.2. The Laplace Block: Hardcoding heat diffusion

As a core component of our physics-hardcoded operator $\mathcal{F}$, this block is designed to accurately and efficiently compute the linear heat diffusion. As depicted in Fig. 1(d1), it is structured as a hybrid model that combines a fixed physical backbone (Discrete Laplacian) with a lightweight neural corrector. The rationale for this design is to directly hardcode well-defined heat diffusion laws to ensure model stability and physical consistency, while simultaneously using a lightweight neural network to learn and correct for potential geometric errors introduced by mesh discretization. This architectural paradigm, which decomposes a physical system's operator into a deterministic, physics-based component and a learned, data-driven residual, is inspired by recent successful frameworks in computational fluid dynamics simulation [28].

Physical Backbone: Discrete Laplacian

The core of this physical backbone is the non-learnable, sparse Laplacian matrix **L**, which was established in Section 2.4.1 from the discrete Laplace–Beltrami operator. As shown in the lower branch of Fig. 1(d1), its primary role is to perform a highly efficient sparse matrix–vector product, $\Delta \mathbf{T} \approx \mathbf{LT}$, to deterministically compute the linear diffusion process. This single, physics-based operation directly injects the fundamental law of heat diffusion (the second law of thermodynamics) into the model, thereby providing a robust, reliable, and training-free baseline while ensuring both physical consistency and numerical stability.

Neural Corrector: A Nonlinear Enhancement to the Discrete Laplacian

Complementing the linear physical backbone, the Neural Corrector is a Message Passing Neural Network (MPNN) designed to enhance the classical discrete Laplacian. The specific architecture of the MPNN, which involves an encode-process-decode workflow as shown in Fig. 3, is detailed in the GNN Block section. Unlike the main GNN block which learns thermodynamic evolution, the Neural Corrector is engineered strictly as an operator-level compensator.

It is mathematically established that the discrete Laplace–Beltrami operator exhibits inherent approximation errors and systematic truncation biases on unstructured finite grids [29,30], which cannot be resolved simply by increasing node count due to local geometric irregularities. Therefore, the sole role of the Neural Corrector is to learn and compensate for these mathematically inherent geometric imperfections. To achieve this, as depicted in the upper branch of Fig. 1(d1), the MPNN uses strictly geometric features as direct inputs, including spatial coordinates $\mathbf{x}_i$, Euclidean distances $\|\mathbf{x}_i - \mathbf{x}_j\|$, and directional vectors $\mathbf{x}_j - \mathbf{x}_i$, while explicitly excluding the temperature variables **T**.

This critical design choice ensures the module isolates spatial discretization errors, completely avoiding redundancy with the main GNN Block. The computed geometric correction term is then added to the output of the physical backbone(**LT**). The final output of the Laplace Block is therefore this sum, scaled by the thermal diffusivity α:

$$\alpha(\mathbf{LT} + \text{Correction}_{NN}(\mathbf{x}_i, \|\mathbf{x}_i - \mathbf{x}_j\|, \mathbf{x}_j - \mathbf{x}_i)) \tag{14}$$

which represents an augmented, high-fidelity approximation of the Laplacian operator. This enhanced operator is then used to compute the diffusive part of the system's time derivative.

2.4.3. The GNN Block: Learning nonlinear residuals

Complementing the Laplace Block, the GNN Block's core task is to learn and approximate all complex nonlinear dynamics that the linear diffusion model cannot capture. These phenomena, which are difficult to model with a single analytical formula, include spatio-temporal effects from the moving laser source, nonlinear boundary conditions like radiation, and temperature-dependent material properties. To this end, and as visualized in Fig. 1(d2), we employ a standard Encode-Process-Decode architecture. Its workflow is designed to transform physical information into learnable features, perform nonlinear evolution in the feature space, and finally map the results back to the physical domain.

Encoder: The Encoder's role is to map the system's raw physical state and geometry, defined as node features $\mathbf{h}_i^n$ and edge features $\mathbf{e}_{ij}^n$ in Section 2.3.1, into high-dimensional latent vectors suitable for graph-based learning. Using distinct Multi-Layer Perceptrons (MLPs), it processes nodes and edges separately: for each node, it fuses the current temperature with its spatial position; for each edge, it fuses the temperature difference between connected nodes with their geometric relationship. This step creates rich initial state representations that embed the critical physical and spatial information necessary for the subsequent processing stages.

Processor: The Processor is the computational core of the GNN Block, responsible for learning the complex, non-local interactions within the system. It consists of multiple stacked layers of a standard GNN variant called MPNN layers [31–33], whose internal structure is

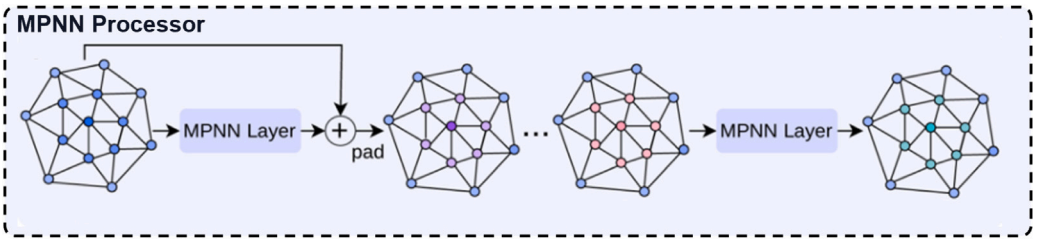


Fig. 3. Detailed architecture of the MPNN processor. The processor consists of multiple stacked MPNN layers. In each layer, node latent representations are iteratively refined by aggregating messages from their topological neighbors. Crucially, additive residual connections (+ pad) are employed between sequential layers to facilitate stable gradients and effectively learn complex, long-range nonlinear spatial dependencies.

detailed in Fig. 3. In each layer, a node's latent representation is iteratively refined by aggregating information from its neighbors. Stacking these layers allows information to propagate across the entire mesh, enabling the model to capture the intricate spatial dependencies that define the nonlinear phenomena. Crucially, residual connections are employed between layers. This design choice not only ensures stable training for a deep network but also enables each layer to effectively learn a correction to the output of the previous one.

Decoder: The Decoder's task is to translate the final, highly-refined latent vectors learned by the Processor back into a concrete, physically meaningful quantity. It takes the latent vector from each node and passes it through a final MLP. This projection maps the complex, learned features from the high-dimensional space down to a single scalar value: the predicted rate of temperature change due to the nonlinear residual. This data-driven prediction is then directly added to the output of the Laplace Block to form the complete instantaneous rate of change for the system.

2.4.4. Incorporating external source and boundary terms in $\mathcal{F}$

Having defined the operator components for the system's internal dynamics, we now incorporate the external physical effects: the moving heat source and boundary heat losses. These terms are modeled as explicit, deterministic additions to the operator $\mathcal{F}$.

Moving Heat Source: The moving laser beam acts as a localized, non-stationary external heat source, modeled as a time-varying surface heat flux [34]. We adopt a 2D Gaussian distribution to describe the flux intensity, which moves along the positive x-axis at a constant scan speed. The flux at a point (x, y) on the top surface can be expressed as:

$$Q_{src}(x, y, t) = Q_{src}(t) = \frac{2\eta P}{\pi r_b^2} \exp\left(-\frac{2((x - v_s t)^2 + y^2)}{r_b^2}\right) \tag{15}$$

where (x, y) are the coordinates on the surface relative to a starting point, assuming the scan starts at the origin and moves along the $y = 0$ line. Here, η is the material's absorptivity, P is the laser power, r_b is the effective beam radius, and v_s is the constant scan speed.

In our graph-based framework, this surface flux is applied to nodes near the top surface, resulting in a discrete, time-varying heat source sequence $Q_{src}^{(1)}, \ldots, Q_{src}^{(n)}$ for all nodes n over time t, which is pre-computed and stored in the dataset. This pre-computed term, which corresponds to the 'Heat Source' input in Fig. 1(d3), is non-dimensionalized and incorporated as a direct additive component into the expression for the final operator.

Boundary Conditions: Complementing the heat source, boundary conditions govern thermal exchange with the surroundings. Convective and radiative heat losses at the surface are modeled as Robin-type boundary conditions and are expressed non-dimensionally using the Biot (Bi) and Stark (St) numbers [35,36]. These heat loss fluxes, computed only for nodes on the relevant boundaries, are incorporated as explicit subtractive terms within the operator $\mathcal{F}$. This approach, represented by the 'Boundary Flux' component in Fig. 1(d3), elegantly integrates boundary physics directly into the model's dynamics.

2.4.5. The complete physics-hardcoded operator

The transient evolution of the temperature field is driven by the interplay of all previously described physical effects. Our framework unifies them into a single, complete operator $\mathcal{F}$ that computes the system's instantaneous rate of change, $\frac{dT}{dt}$. As summarized by the final equation in Fig. 1(d), this operator is the sum of the four core components:

$$\mathcal{F} = \underbrace{\alpha \cdot \text{Laplace_Block}(\mathbf{T})}_{\text{Internal Dynamics}} + \underbrace{\text{GNN_Block}(\mathbf{T})}_{\text{Nonlinear Residual}} + \underbrace{Q_{src}^{(n)}}_{\text{Source}} - \underbrace{\left[\text{Bi}(T - T_{amb}) + \text{Sk}(T^4 - T_{amb}^4)\right]}_{\text{Boundary Flux}} \tag{16}$$

This hybrid structure inherently guides the learning process during optimization. The analytical components (Laplace Block, sources, boundaries) provide a robust physical scaffold, freeing the learnable GNN Block to focus exclusively on modeling the complex, nonlinear residual dynamics. The result is a synergistic framework where the interpretable physics-hardcoded parts and the powerful data-driven component complement each other, achieving both high accuracy and physical consistency. This complete operator is then passed to a high-order numerical integrator to advance the system in time, as detailed in the next section.

2.5. Temporal integration for transient simulation

To capture the transient evolution of the temperature field, the system of ODEs derived in Section 2.3.2 must be advanced in time. Given the long-duration, multi-thousand-step nature of MAM thermal simulations, low-order methods like the Forward Euler scheme are prone to accumulating significant error, potentially leading to numerical instability [37].

To ensure both stability and accuracy, we couple our physics-hardcoded operator with a higher-order numerical integrator. Specifically, we employ a second-order Runge–Kutta method (RK2), specifically Heun's method, to perform the time-stepping. Each full-time step from t_n to $t_{n+1} = t_n + \Delta t$ involves:

Predictor Step: First, an intermediate state $T_{n+1}^{(1)}$ is estimated using the derivative at the current time:

$$\begin{aligned} K_1 &= \mathcal{F}(T_n, t_n) \\ T_{n+1}^{(1)} &= T_n + \Delta t \cdot K_1 \end{aligned} \tag{17}$$

Corrector Step: The derivative is then re-evaluated at the estimated future state, and the two derivatives are averaged to compute the final update:

$$\begin{aligned} K_2 &= \mathcal{F}(T_{n+1}^{(1)}, t_n + \Delta t) \\ T_{n+1} &= T_n + \Delta t \cdot \frac{(K_1 + K_2)}{2} \end{aligned} \tag{18}$$

Crucially, each evaluation of the operator $\mathcal{F}(\cdot)$ corresponds to a full forward pass of our dual-branch GNN architecture, as visualized

Table 1
Particle size distribution for Inconel 718 powder.

Diameter (μm)	Concentration (%)
20	30
25	40
30	15
35	10
40	5

Table 2
Thermophysical properties of Inconel 718.

Property	Value	Unit
Liquidus temperature	1608.15	K
Solidus temperature	1523.15	K
Latent heat of fusion	210,000	J/kg
Density	7401.97	kg/m^3
Thermal conductivity	29.598	W/m K
Specific heat capacity	720	J/kg K
Surface tension coefficient	1.882	N/m
Latent heat of vaporization	6.6×10^6	J/kg
Reference temperature	1608.15	K

in Fig. 1(e). This design elegantly embeds the learned physics model directly within the numerical integration scheme. By approximating the rate of change at each sub-step, the influences of diffusion, nonlinear residuals, and external heat sources are naturally incorporated into the state update. For applications demanding even higher fidelity, this framework can be seamlessly extended to use integrators like the fourth-order Runge–Kutta (RK4) [38].

3. Experimental validation

To validate the performance of the proposed MAM-PhyGNN framework, this section details a challenging case study focused on the mesoscopic thermal dynamics of LPBF. We selected the LPBF of Inconel 718 as a representative test case due to its industrial relevance and complex physical behavior. The primary objective is to assess the model's ability to accurately and stably predict the powder-scale thermal field in long-term autoregressive simulations, thereby demonstrating its potential as a fast and reliable surrogate for high-fidelity solvers.

3.1. High-fidelity simulation for dataset generation

To generate a ground-truth dataset for training and validation, we conducted a series of high-fidelity transient thermal simulations using the commercial software FLOW-3D AM 2025R1. This section details the physical modeling, simulation parameters, and the subsequent data preprocessing pipeline used to construct the graph-based dataset.

3.1.1. Physical modeling and simulation parameters

To isolate the core thermal physics, fluid dynamics were intentionally neglected, and a transient heat conduction model was employed [39]. The simulation domain was a rectangular region of 200 μm × 100 μm × 50 μm, discretized with a uniform Cartesian mesh of 10 μm cell size. A powder bed with an approximate equivalent layer thickness of 80 μm was generated within this domain using the software's integrated Discrete Element Method (DEM) model. The powder material was Inconel 718, with a particle size distribution detailed in Table 1. The key thermophysical properties of Inconel 718 used in the simulation are summarized in Table 2.

A critical component of the physical model is the laser energy absorption. Recognizing that absorptivity is a dynamic variable dependent on local geometry rather than a fixed constant, a Fresnel absorption model was employed to calculate it dynamically. This yields more physically accurate temperature field data. Each simulation was executed for a total duration of 2.4 ms. The complete temperature field was recorded at intervals of 0.0012 ms, resulting in a sequence of 2000 data frames for each simulation case. A representative top-down view of the simulated temperature field at $t = 1.2$ ms (frame 1000), for a case with 300 W laser power and 800 mm/s scan speed, is illustrated in Fig. 4.

Table 3
Dataset partitioning and process parameters.

Dataset partition	Case	Laser power (W)	Scan speed (mm/s)
Training set	Case1	300	1000
	Case2	300	800
	Case3	300	1200
	Case4	100	1000
Test set	Case5	200	1000

3.1.2. Dataset construction and partitioning

The simulation process described in Section 3.1.1 was repeated to generate five distinct cases with varying process parameters, as summarized in Table 3. To ensure a robust evaluation of generalization, four cases were allocated to the training set, while the remaining case was reserved for testing. This strict separation ensures that the model is evaluated on entirely unseen data, providing a reliable assessment of its generalization capability across different process conditions.

The raw simulation data required careful preprocessing to be converted into a format compatible with our GNN architecture. This transformation involved two main steps: graph construction and temperature field mapping, a process visually summarized in Fig. 5.

First, the unstructured computational graph was constructed. A set of nodes was generated via adaptive Poisson disk sampling within the 2D domain [40], as illustrated in Fig. 5(a). This strategy places a higher density of nodes in critical regions, such as along the laser path, while using a sparser distribution elsewhere to maintain computational efficiency. Specifically, this sampling strategy resulted in a highly sparse graph consisting of exactly 2236 points. Subsequently, a Delaunay triangulation was performed on this point set to generate the final computational mesh [41]. This mesh defines the graph's topology, where nodes are the sampled points and edges represent the connections between them. As shown in Fig. 5(b), this methodology produces well-shaped triangular elements, a critical feature that enhances the numerical stability and accuracy of the discrete Laplace–Beltrami operator used later by preventing the occurrence of extreme values in the cotangent weights.

Second, the dynamic temperature data was mapped onto the graph. For each time step, the 3D temperature fields from the FLOW-3D simulations (exported in VTK format) were projected onto the 2D plane of our graph nodes. This was achieved using a vertical ray casting method, an established technique from computer graphics [42]. This technique maps the uppermost temperature value from the 3D data to its corresponding 2D node location, effectively transforming the volumetric data into a surface temperature map. Rather than relying on simplified 2D simulations, this projection strategy is deliberately chosen to implicitly embed critical 3D physical mechanisms (e.g., volumetric laser-powder interactions) into the dataset. Concurrently, it mimics the field of view of industrial coaxial infrared cameras, ensuring the model's direct applicability for in-situ monitoring and digital twin integration. Ultimately, this process yields a sequence of nodal temperatures, creating a dynamic graph signal for training and evaluation. The result of this mapping at a single time step is visualized in Fig. 5(c), which shows the smooth, interpolated temperature contours, and Fig. 5(d), which displays the final discrete temperature values assigned to each node, representing the final input signal for the GNN model.

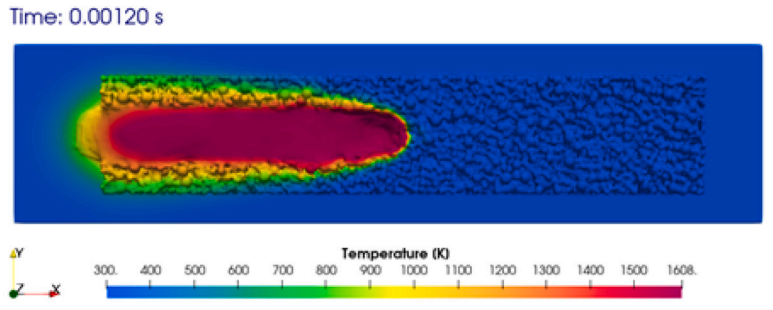


Fig. 4. Top-down view of the simulated temperature field at $t = 0.0012$ s for a case with 300 W laser power and 800 mm/s scan speed.

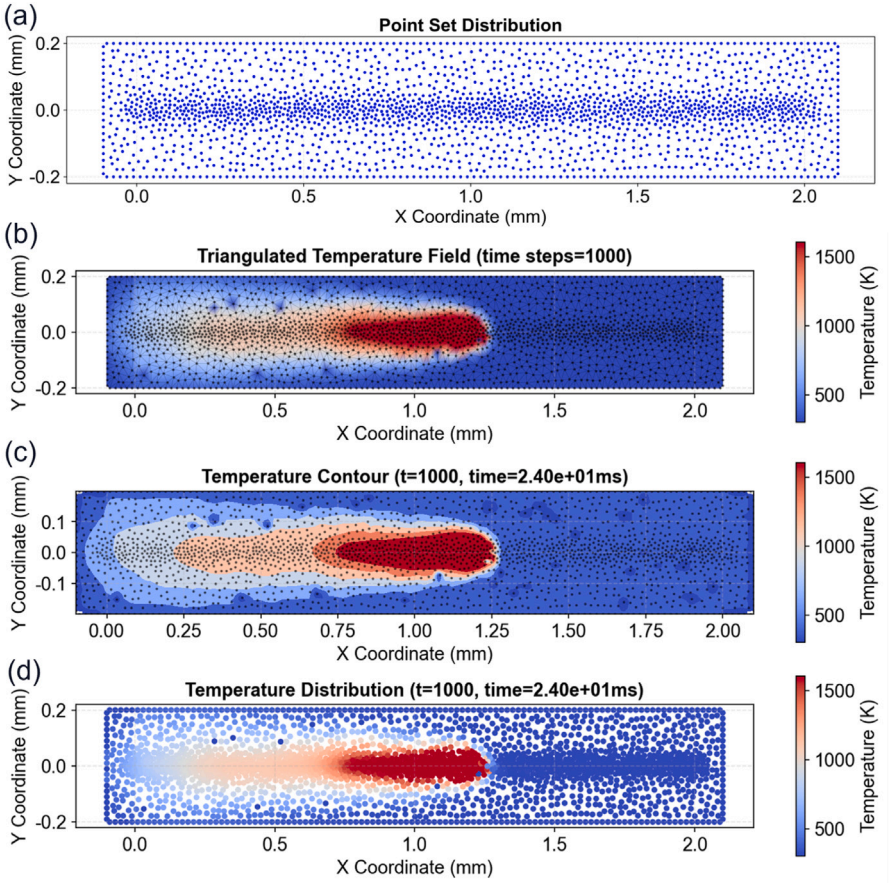


Fig. 5. Graph construction and data mapping pipeline. (a) The distribution of nodes in the 2D computational domain, generated via adaptive Poisson disk sampling. (b) The final triangulated mesh overlaid with the interpolated temperature field at a single time step ($t = 1000$). (c) The corresponding smooth temperature contours, highlighting the shape of the melt pool and heat-affected zone. (d) The discrete temperature values assigned to each individual node, representing the final input signal for the GNN model.

3.2. Model configuration and training details

This section details the specific configuration of MAM-PhyGNN and the parameters used for its training and evaluation.

3.2.1. Non-dimensionalization parameters

As described in Section 2.1, the governing equations and all associated variables were non-dimensionalized to improve numerical stability and promote model generalization. The key characteristic scales and derived dimensionless parameters used in this study are summarized in Table 4.

Table 4
Non-dimensionalization parameters used in this case.

Parameter	Symbol	Value	Unit
Characteristic length	L_{ref}	200	μm
Characteristic temperature	T_{ref}	1610	K
Reference thermal diffusivity	α_{ref}	2.29×10^{-6}	m^2/s
Characteristic time	t_{ref}	17.5	ms

3.2.2. Network architecture

The architecture of the MAM-PhyGNN is a direct implementation of the dual-branch operator described in Section 2.4. The specific

Table 5
MAM-PhyGNN model configuration for the LPBF case study.

Component	Sub-Component	Parameter	Value
Global	Temporal integrator	Method	2nd-Order Runge–Kutta (RK2) Advances state from t to $t + \Delta t$
	Optimizer	Name	AdamW
		Learning rate	1e−4
		Training epochs	200
Laplace Block	Physical backbone	Operator	Discrete Laplace–Beltrami (L) Pre-computed and fixed; based on mesh geometry.
	Neural Corrector	Architecture	MPNN
		Input features	Geometric only (edge length, etc.) Excludes temperature to prevent redundancy.
		Hidden dimension	32 A lightweight network for discretization error correction.
GNN Block	Encoder	Node encoder	MLP [3, 32, 64, 64] Input: 2D coords + 1D temp.
		Edge encoder	MLP [4, 32, 64, 64] Input: 3D edge attr. + 1D temp. diff.
	Processor	Architecture	5 stacked MPNN With residual connections.
		Message MLP (ϕ)	[192, 64, 64] Input size = 2*(NodeEmb) + EdgeEmb = 2*64 + 64 = 192.
		Update MLP (γ)	[128, 64, 64] Input size = NodeEmb + AggregatedMsg = 64 + 64 = 128.
	Decoder	Decoder	MLP [64, 32, 16, 8, 1] Projects final node embedding to a scalar rate of change.

hyperparameters for the LPBF case study are summarized in Table 5. The configuration strategically instantiates the framework, with the Laplace Block's neural corrector restricted to geometric inputs to prevent redundancy, and a deep, five-layer processor in the GNN Block to capture non-local melt pool dynamics.

The configuration detailed in Table 5 strategically instantiates the MAM-PhyGNN framework for the LPBF problem. The dual-branch architecture is central to this design. Notably, the Laplace Block's neural corrector is intentionally restricted to geometric inputs, ensuring that it only corrects for discretization errors while the non-learnable backbone strictly enforces the fundamental diffusion physics. Complementing this, the GNN Block's five-layer deep processor is designed to have a sufficiently large receptive field to capture the complex, non-local interactions characteristic of melt pool dynamics. This synergistic division of labor, combined with a stable RK2 time integrator, creates a model that is both physically grounded and flexible enough to learn the intricate nonlinearities of the LPBF process.

For training, we employed the AdamW optimizer with a StepLR scheduler to ensure both rapid initial convergence and stable fine-tuning. A time stride of 9 was used during data sampling to encourage the model to learn longer-term temporal dependencies. The loss was computed as the Mean Squared Error across a 10-timestep prediction window, promoting stable and accurate long-term predictions. Benefiting from the computational efficiency of the physics-hardcoded architecture, all experiments were conducted using a single NVIDIA A100-80GB GPU.

3.3. Key results and analysis

3.3.1. Short-term prediction performance

To comprehensively evaluate the model's generalization capability on out-of-distribution data and its stability in long-term forecasting, the model trained on case1 through case4 was deployed to predict the temperature field evolution of an entirely unseen test set, case5. This section assesses the model's spatiotemporal prediction performance, beginning with a qualitative validation of its generalization, followed by a detailed quantitative analysis of its predictive accuracy.

Before presenting the results, it is important to clarify the terminology used in our time-series analysis. The full data from each simulation consists of a long sequence of frames, where each frame's position is its actual time step (e.g., from 0 to 1999). For training and evaluation, this sequence is sampled to create training instances, a process governed by three key parameters: window size, time stride, and window step, as illustrated in Fig. 6(a).

First, a window size of 10 defines that each training instance consists of 10 data points. However, these points are not consecutive frames. Instead, they are sampled with a time stride of 9. This means that within a single training window, the included frames are spaced 9 actual time steps apart (e.g., frames at actual time steps {0, 9, 18, . . . , 81}). This sparse sampling strategy is specifically designed to train the model to predict system states over a much longer effective time horizon (in this case, 90 time steps) from one step to the next, thereby encouraging it to learn the underlying long-term physical dynamics rather than just short-term correlations. Finally, these windows themselves are generated with a window step of 10, meaning the starting point of each new window is 10 actual time steps after the start of the previous one. In our autoregressive predictions, the task is to predict the full sequence of 10 sparsely sampled frames within a window, using only the first frame (at local $t = 0$) as the initial condition. A "10-step forecast" thus refers to recursively predicting the states at local time steps $t = 0$ through $t = 9$.

The model's generalization across different stages of the AM process is assessed visually in Fig. 6(b). Four distinct time windows (Window 10, 50, 100, and 150) were selected, each representing a different phase of the simulation. For each window, a 10-step autoregressive forecast was performed, and the final predicted frame (at local $t = 9$) was compared to its corresponding ground truth. This final frame corresponds to a state 81 actual time steps after the initial condition, demonstrating the model's ability to forecast over a significant time horizon. The results clearly demonstrate that the predicted temperature fields maintain a high degree of fidelity with the ground truth in terms of melt pool morphology, size, and the overall shape of the heat-affected zone (HAZ), regardless of the process stage. The corresponding error heatmaps for each window explicitly visualize the temperature deviations. As observed, the localized errors are predominantly confined to the immediate boundaries of the moving melt pool, which are regions characterized by the steepest spatial and temporal temperature gradients. Meanwhile, the absolute error magnitudes remain remarkably low across the global domain. This confirms that the physical laws learned by the model generalize exceptionally well and can accurately capture dynamic evolutionary features over time.

A more detailed quantitative and qualitative analysis of a single representative forecast is presented in Fig. 7. The temporal evolution

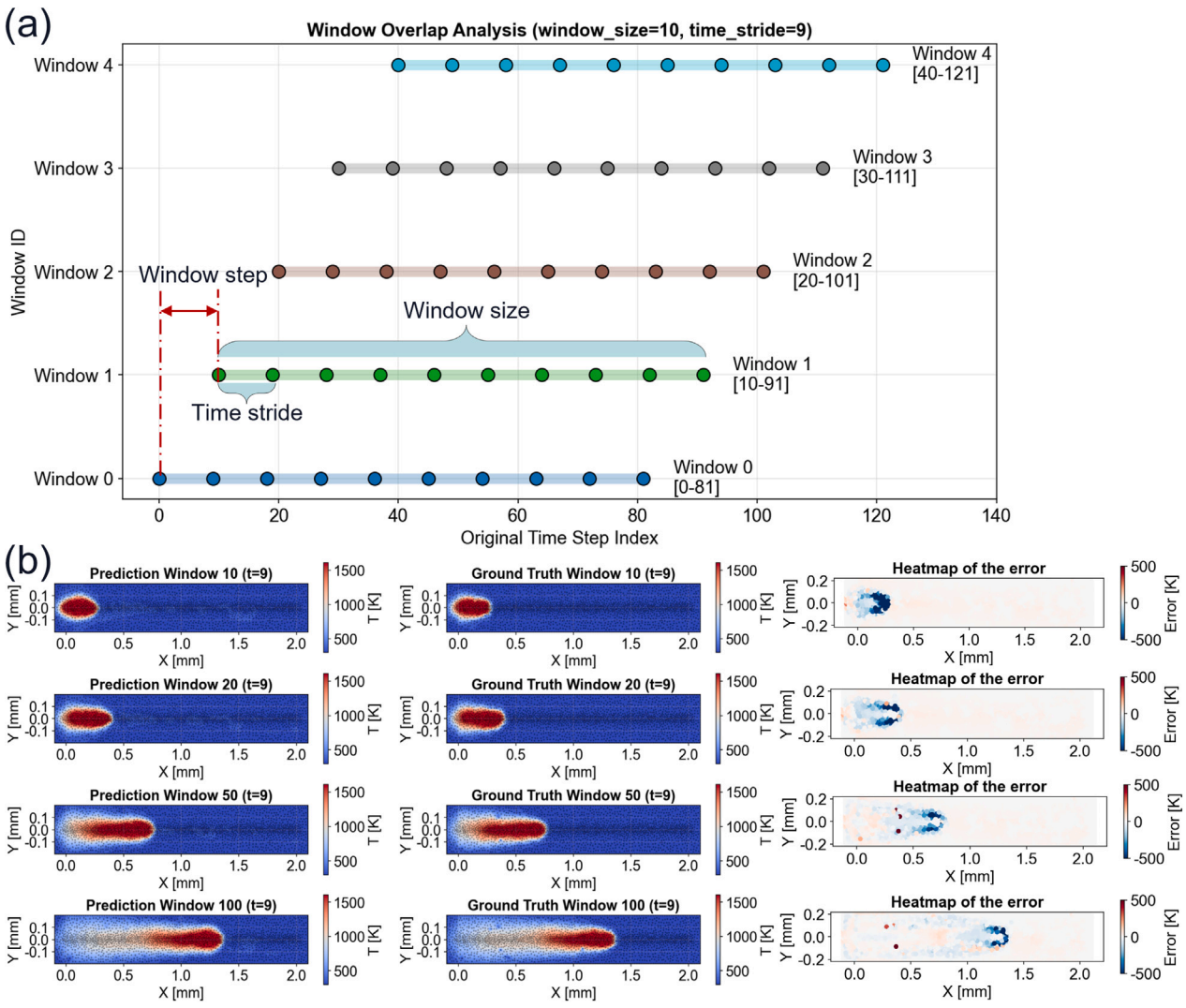


Fig. 6. Qualitative analysis of autoregressive prediction performance on an unseen test case. (a) Schematic of the overlapping sliding window strategy used for data sampling, with a window size of 10 and a time stride of 9. (b) Visual comparison of the predicted final frame ($t = 9$) versus the ground truth, along with their corresponding spatial error distribution heatmaps for four different time windows, demonstrating the model's generalization capability across various process stages.

within this window, shown in Fig. 7(a), displays the model's predictions at intermediate steps (local $t = 3$, $t = 6$, and $t = 9$) against their ground truth counterparts. The consistent agreement at each stage validates that the model not only predicts the final state accurately but also simulates the intermediate evolutionary path with high fidelity.

The quantitative error analysis for this 10-step forecast is detailed in Fig. 7(b). The top-left panel, showing temporal error accumulation, reveals a monotonic increase in Mean Absolute Error (MAE) with each prediction step, illustrating the inherent error accumulation in autoregressive forecasting. Concurrently, the widening of the standard deviation band indicates that prediction uncertainty grows with the forecast horizon. The top-right panel displays the spatial distribution of error at the final prediction step (local $t = 9$). The error is not randomly distributed; instead, localized errors are concentrated in regions with steep temperature gradients, such as the leading and trailing edges of the melt pool. This is an expected outcome, as these areas represent the most physically complex and dynamic zones of the simulation.

In summary, these analyses demonstrate that the model exhibits both strong generalization capabilities and robust predictive performance on unseen test data. It accurately reproduces the spatiotemporal evolution of the temperature field, even when forecasting over extended, sparsely-sampled time horizons. The observed error characteristics are consistent with the known challenges of autoregressive modeling and highlight the physically demanding regions of the simulation, providing clear insights for future work. These results validate the model's performance for short-term, single-window forecasts. The following section will test its stability over a much longer, continuous prediction horizon.

3.3.2. *Long-term autoregressive prediction performance*

While Section 3.3.1 validated the model's excellent performance within a single prediction window (10 steps), this section subjects the model to a more rigorous test of its long-term stability. We evaluate this through an extended autoregressive prediction mode. This procedure is initiated by providing the model with a single ground truth temperature field as the initial condition ($t = 0$). Subsequently, the model operates in a closed-loop fashion: the prediction for each time step is used as the input for the next, with no further corrections from the ground truth temperature data. This procedure evaluates the model's ability to function as a true forward simulator over a long horizon.

The process is initiated by providing the model with the first full window of ground-truth data. The model then performs a 10-step

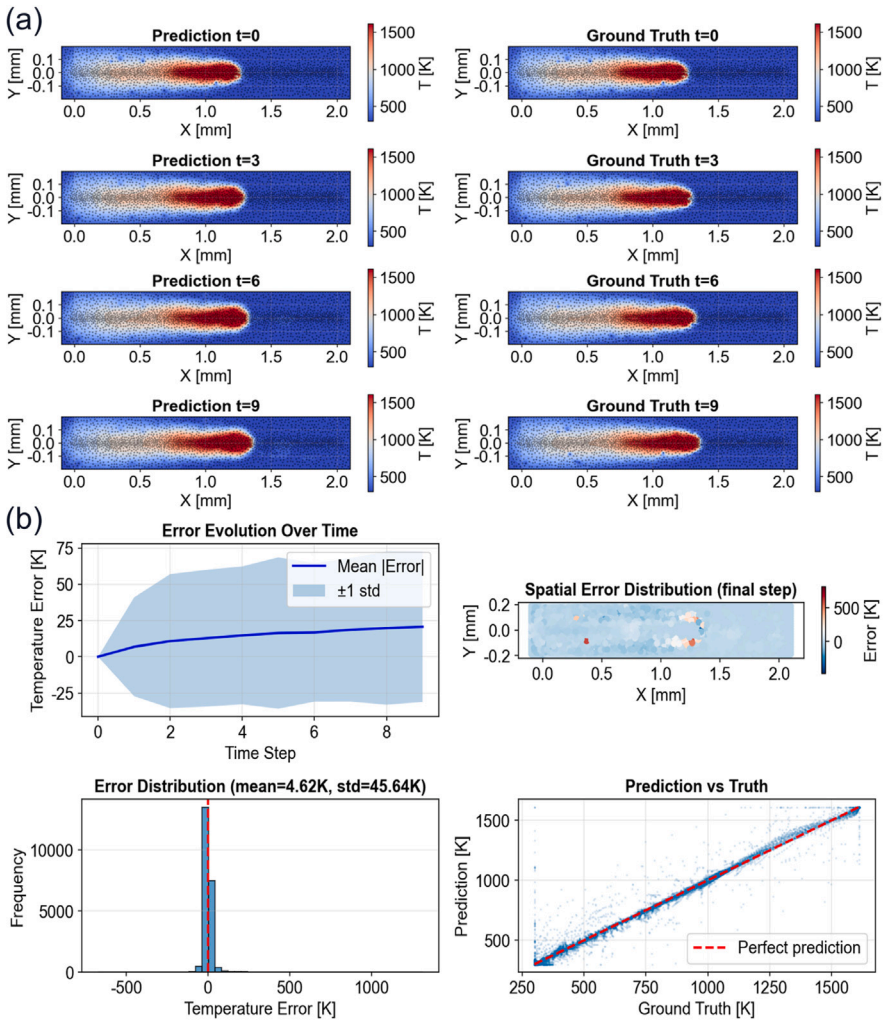


Fig. 7. Quantitative analysis of autoregressive prediction performance on an unseen test case. (a) Temporal evolution of the predicted temperature field within a single window, illustrating the progressive forecast at steps $t = 3$, $t = 6$, and $t = 9$ against their corresponding ground truth frames. (b) Comprehensive quantitative error analysis of a 10-step autoregressive prediction, including: (top-left) mean absolute error and its standard deviation evolving over time; (top-right) spatial distribution of the temperature error at the final prediction step; (bottom-left) histogram of temperature errors (K) across all spatiotemporal points; and (bottom-right) a scatter plot comparing predicted values to ground truth values (K), where the dashed line represents a perfect prediction.

autoregressive forecast to predict the entire window's evolution. Crucially, the final predicted frame of this window is then used as the initial condition for the next, completely non-overlapping window, as illustrated in the schematic in Fig. 8(a). This closed-loop process, where the model's own long-range forecast becomes the input for the subsequent long-range forecast, is repeated sequentially for 10 windows, covering almost 1000 actual time steps. This non-overlapping long-term prediction is a significantly more challenging test than single-window prediction. Any error from one window's forecast is directly propagated as the initial error for the next, making the test extremely sensitive to error accumulation. Therefore, success in this task is a powerful demonstration of the model's ability to maintain physical consistency and avoid divergence over long horizons, a key metric for its potential as a digital twin.

The qualitative results of these long-term predictions are presented in Fig. 8(b). It provides a visual comparison of the model's predictions against the ground truth at various points during the 981-step prediction. Snapshots are shown at prediction steps $t = 19$, $t = 29$, $t = 39$, $t = 49$, and $t = 99$ relative to the start of the forecast. Even after nearly 1000 steps (corresponding to actual time step 981), it is evident that the model accurately captures the dynamic elongation of the melt pool as the laser source moves. The predicted melt pool morphology, size, and peak temperature zones remain remarkably consistent with the ground truth. The corresponding error heatmaps illustrate the spatial evolution of the prediction error over this extended horizon. As the autoregressive prediction advances, the accumulated errors become visible, predominantly localized at the dynamic boundaries of the melt pool and along its trailing wake. Although a minor blurring effect emerges at the tail of the melt pool in later stages due to this expected error accumulation, the heatmaps indicate that the deviations are primarily confined to these specific regions. The overall thermal structure remains intact, which supports the model's long-term stability.

To precisely evaluate this long-term performance, a comprehensive quantitative analysis was conducted, with results presented in Fig. 9. The top panels of Fig. 9(a) illustrate the temporal error evolution. Both the MAE and the mean relative error exhibit a slow, controlled growth, reaching approximately 90 K and 10%, respectively, after nearly 1000 time steps. Crucially, the growth trend is sub-linear and bounded, showing no signs of exponential explosion or divergence, which provides strong evidence of the model's long-term stability. The gradual widening of the standard deviation bands (shaded areas) also aligns with the physical intuition that uncertainty increases over longer forecast horizons.

The bottom panels of Fig. 9(a) further dissect the overall error characteristics. Although the standard deviation (93.45 K) and mean (30.13

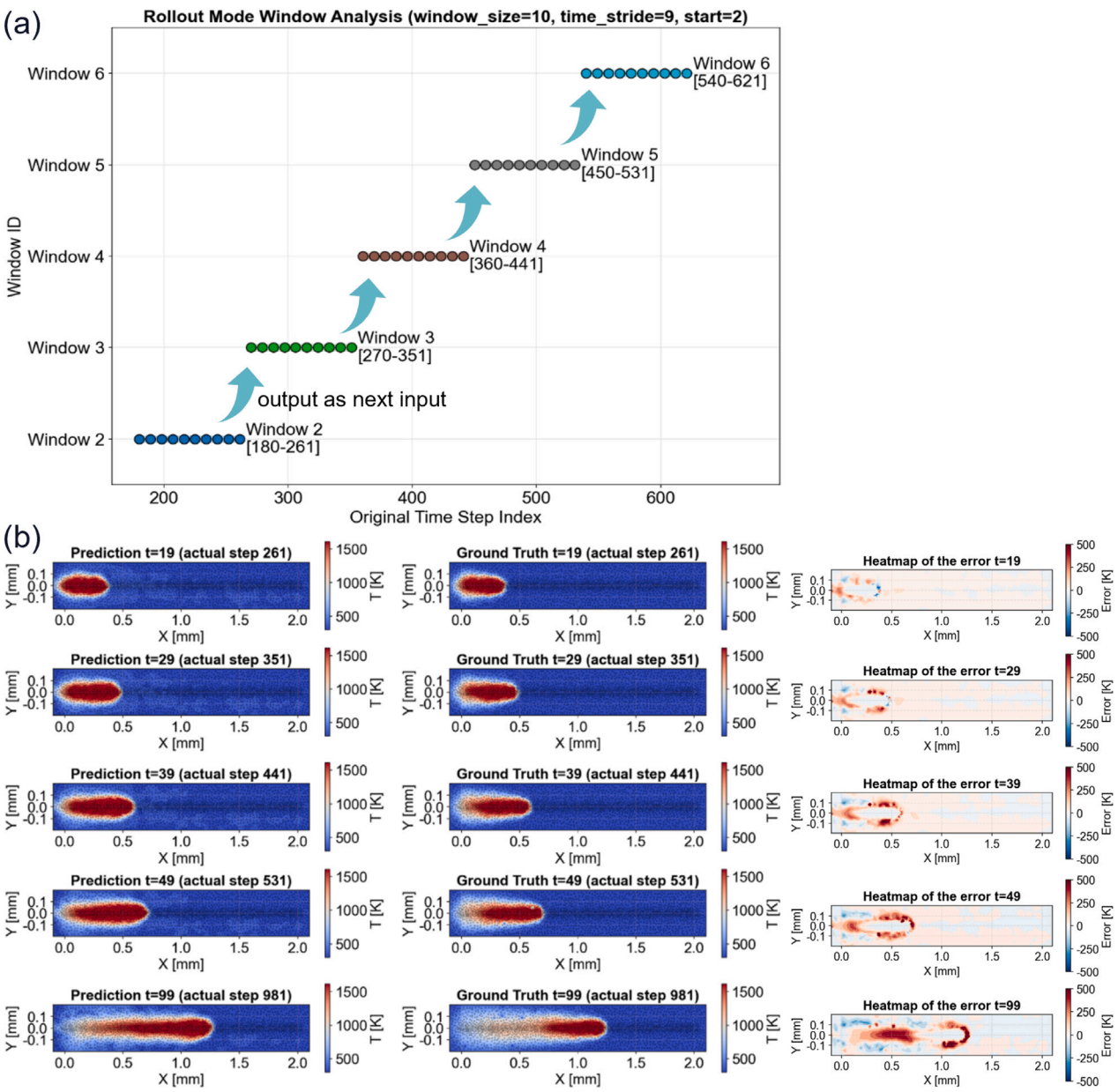


Fig. 8. Qualitative analysis of long-term autoregressive rollout performance on an unseen test case. (a) Schematic illustrating the long-term prediction process, where the model recursively forecasts over 100 consecutive windows. (b) Qualitative comparison of predicted temperature fields versus ground truth at selected time steps during a 981-step long-term prediction process, demonstrating the model's ability to capture long-term dynamics.

K) of the error are larger than those in the single-window prediction, the error histogram still displays a sharp peak centered around zero, indicating that the majority of predictions remain accurate. Similarly, the prediction-vs-truth scatter plot, while more dispersed than its short-term counterpart, shows that data points still cluster tightly around the ideal diagonal line, confirming a strong linear correlation. Finally, Fig. 9(b) offers the most compelling evidence of the model's structural and statistical stability. This plot shows the Pearson correlation coefficient between the predicted and ground truth fields at every time step. A gradual decline in correlation is observed as the autoregressive prediction advances, reflecting the expected error accumulation over long horizons. Nevertheless, the overall correlation remains exceptionally high, staying above 0.96 even approaching the 1000th actual time step with a final global value of 0.9803. This indicates that despite the inevitable accumulation of absolute errors, the spatial structure and complex dynamic patterns of the predicted temperature field are well preserved.

Collectively, the results from this pure autoregressive prediction unequivocally demonstrate the model's outstanding long-term prediction stability. The controlled growth of error and the consistently high correlation coefficient prove that the model has successfully learned and internalized the underlying physical evolution laws.

3.3.3. *Ablation study*

Having demonstrated the robust long-term stability of the full MAM-PhyGNN model in the preceding sections, we now conduct a rigorous ablation study to dissect the architecture and quantify the contribution of its core components. The central hypothesis of our work is that the synergistic combination of a data-driven GNN and a physics-informed operator is essential for achieving both accuracy and long-term stability. To test this, we evaluate three ablated variants against our full model on the same demanding 100-step autoregressive prediction task:

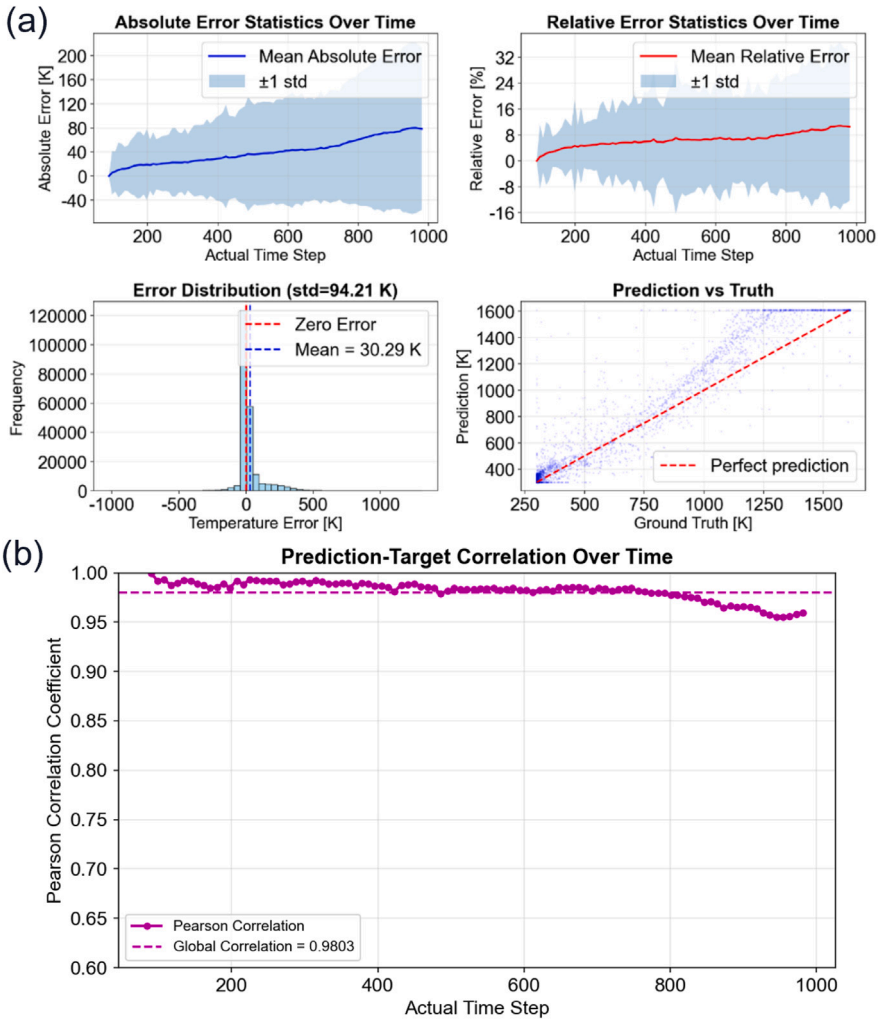


Fig. 9. Quantitative analysis of long-term autoregressive prediction performance on an unseen test case. (a) Comprehensive quantitative error analysis of the entire prediction, showing (top-left) absolute error over time, (top-right) relative error over time, (bottom-left) a histogram of all temperature errors, and (bottom-right) a scatter plot of prediction vs. truth. (b) Pearson correlation coefficient between the predicted and ground truth fields at each time step, highlighting the model's structural and statistical stability.

1. **GNN Block-only**: This variant completely removes the Laplace Block, forcing the model to operate as a purely data-driven GNN. It must learn the system's dynamics, including heat diffusion, from data alone.
2. **Laplace Block-only**: This variant removes the powerful data-driven GNN Block, relying solely on the discrete Laplace–Beltrami operator and its lightweight neural corrector to predict thermal evolution. This model is heavily constrained by the diffusion equation but it does not have access to the temperature field and thus cannot learn complex, state-dependent dynamics from data.
3. **Full w/o Neural Corrector**: This variant completely removes the Neural Corrector from the Laplace Block, forcing the model to rely solely on the purely linear discrete Laplace–Beltrami operator combined with the main GNN block. This configuration is specifically designed to evaluate whether the MPNN-based geometric correction is redundant when a powerful dynamics-learning GNN is already present.

Qualitative analysis of long-term stability. Fig. 10 provides a striking qualitative comparison of these models against the ground truth at three key moments during the long term prediction ($t = 9$, $t = 49$, and $t = 99$).

- **GNN Block-only**: The purely data-driven variant demonstrates a catastrophic failure in long-term stability. While it mimics the initial state reasonably well ($t = 9$), error accumulation leads to a rapid and complete structural collapse. By $t = 49$, the prediction becomes physically implausible, and by $t = 99$, the temperature field has degenerated into a chaotic and noisy state with no resemblance to the ground truth. This result starkly illustrates the necessity of a physical inductive bias to regularize the learning process and prevent divergence in long-term forecasting.
- **Laplace Block-only**: This physically-constrained model performs reasonably well in the initial stages ($t = 9$), accurately capturing the diffusion around the heat source. However, as the long-term prediction progresses, its inherent limitations become apparent. By $t = 49$ and $t = 99$, as it lacks the powerful, data-driven GNN block to account for nonlinearities, the model fails to capture the complex, advection-like dynamics of the moving melt pool. The predicted melt pool becomes unphysically bulbous and distorted at the front, indicating that the linear diffusion operator alone is insufficient to model the nonlinear phenomena associated with phase change and a moving heat source.
- **Full w/o Neural Corrector**: This variant maintains overall long-term stability and successfully captures the general advection-like dynamics of the moving melt pool, confirming the main GNN's

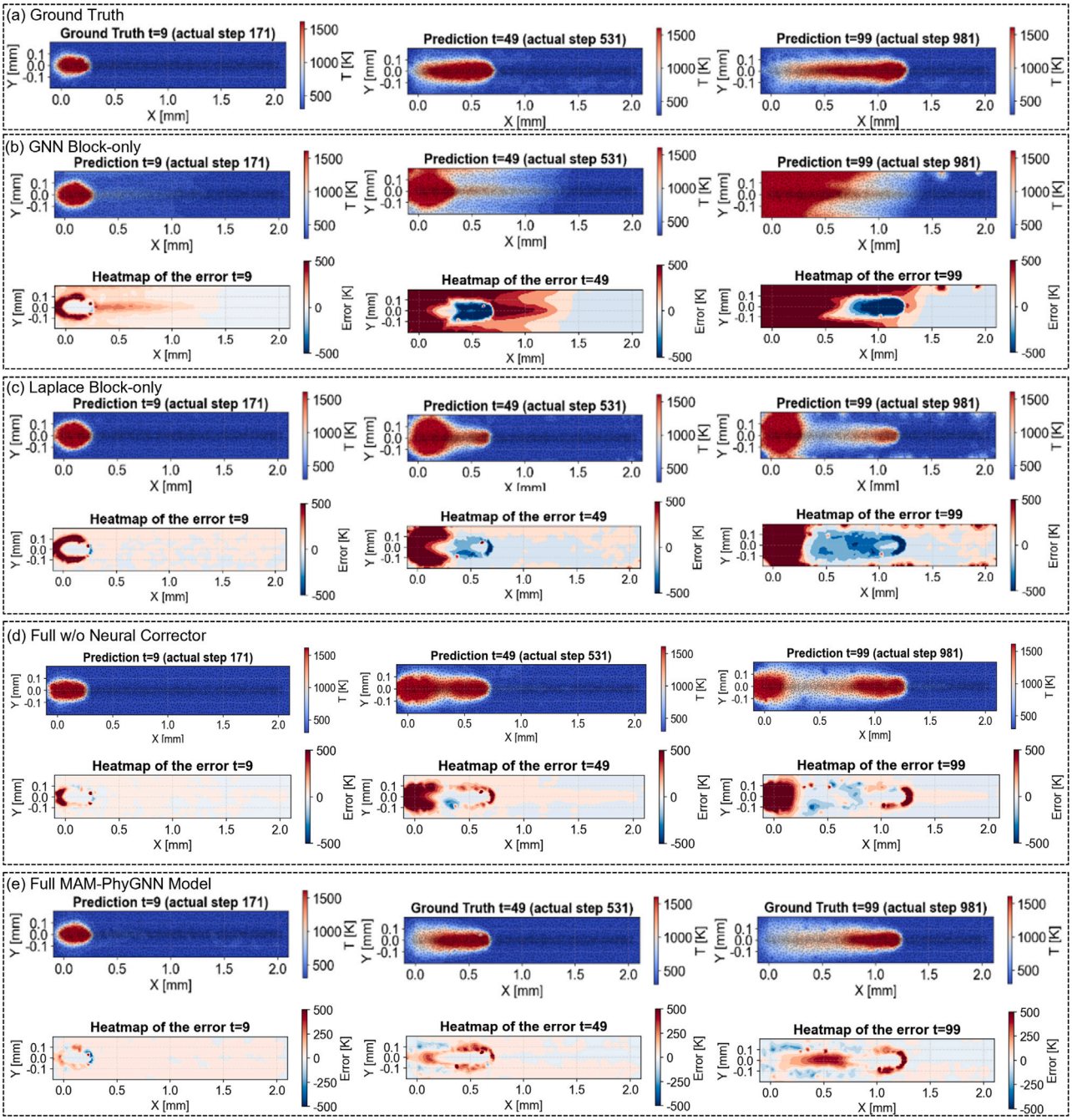


Fig. 10. Qualitative ablation study of long-term rollout stability. The figure provides a visual comparison of predicted temperature fields from the full model and its ablated variants against the ground truth. Snapshots are shown at four key time steps ($t = 0$, $t = 9$, $t = 49$, and $t = 99$) during a 100-step autoregressive prediction to illustrate the long-term stability of each model. The four scenarios visualized are: (a) the Ground Truth from the high-fidelity simulation; (b) the purely data-driven "GNN Block-only" variant; (c) the physics-constrained "Laplace Block-only" variant; and (d) "Full w/o Neural Corrector" variant: the full MAM-PhyGNN model without the Neural Corrector in the Laplace Block; (e) our full MAM-PhyGNN model.

dominant role in modeling nonlinear thermodynamics. However, close visual inspection at later stages ($t = 49$ and $t = 99$) reveals localized artifacts, blurred thermal gradients, and slight morphological deviations compared to the full model. This qualitative degradation visually illustrates how uncompensated spatial truncation errors from the discrete mesh gradually accumulate over time, highlighting the necessity of the geometric correction.

- **Full MAM-PhyGNN Model**: Consistent with the analysis in Section 3.3.2, our full model maintains exceptional fidelity throughout the entire process. The predicted melt pool elongates smoothly, and its thermal footprint remains remarkably consistent with the ground truth, even after 99 autoregressive steps.

Quantitative analysis. The qualitative observations are strongly corroborated by the quantitative metrics summarized in Table 6. The full MAM-PhyGNN model decisively outperforms all ablated variants across all metrics. The GNN Block-only model suffers from the highest error, with its Global RMSE of 409.35 K being over four times higher than that of our full model, and its Pearson Correlation dropping to a poor 0.6758. This confirms that without the stabilizing effect of the

Table 6
Quantitative comparison of the full model and its ablated variants. Best results are in bold.

Model variant	RMSE (K) ↓	MAE (K) ↓	Pearson Corr. ↑	Rel. Err. (%) ↓
GNN Block-only	409.35	258.74	0.6758	57.04
Laplace Block-only	275.04	122.64	0.8276	18.71
Full w/o Neural Corrector	184.69	78.36	0.9195	12.41
Full model (Ours)	**100.45**	**40.91**	**0.9822**	**6.64**

physics-informed Laplace Block, the model is unable to contain error accumulation.

Conversely, the Laplace Block-only model, while significantly better than the pure GNN, still incurs a substantial RMSE of 275.04 K. This demonstrates that while the physical prior is crucial for stability, the data-driven GNN Block is indispensable for learning the complex, nonlinear dynamics that the simple diffusion equation cannot capture.

Furthermore, the evaluation of the newly introduced "Full w/o Neural Corrector" variant highlights the necessity of operator-level geometric compensation. Removing this component causes the RMSE to surge by 83.9% (from 100.45 K to 184.69 K) and the Pearson Correlation to drop from 0.9822 to 0.9195. This substantial performance gap confirms that while the main GNN successfully captures unmodeled thermodynamics, it cannot autonomously resolve the mathematically inherent truncation biases of the discrete mesh. Therefore, the geometric Neural Corrector and the thermodynamic main GNN are highly complementary rather than redundant.

In conclusion, this ablation study unequivocally validates our dual-branch design. The results prove that neither a purely data-driven approach nor a simple physics-constrained model is sufficient for accurate and stable long-term prediction of this complex thermal process. It is the synergistic interplay between the GNN Block's ability to learn nonlinear corrections and the Laplace Block's role as a robust physical regularizer that enables the MAM-PhyGNN framework to achieve its state-of-the-art performance.

3.3.4. Comparative experiment

To further underscore the fundamental advantages of the proposed MAM-PhyGNN framework and ensuring long-term physical consistency, we conducted a direct quantitative and qualitative comparison against a prominent physics-informed surrogate model: the Physics-Informed Neural Network (PINN). Both models were tasked with performing a long-term autoregressive prediction over 100 prediction steps on the same unseen test set (Case 5).

Qualitative analysis: Long-term stability. Fig. 11 provides a striking visual testament to the profound difference in long-term stability between the two models. Our MAM-PhyGNN model demonstrates exceptional stability throughout the long-term prediction. Even at step 99, the predicted melt pool morphology, size, and heat-affected zone remain in high fidelity with the ground truth, proving its robust physical consistency and long-term forecasting capability. In stark contrast, the PINN model suffers a catastrophic failure. Its predictions diverge from the physical reality early on, and by the later stages, the entire temperature field degenerates into physically meaningless noise, with no discernible melt pool structure.

Quantitative analysis: Prediction accuracy. The quantitative metrics in Table 7 further corroborate these qualitative observations, showing that MAM-PhyGNN outperforms the PINN by an order of magnitude across all accuracy metrics. Our model achieves a Root Mean Square Error (RMSE) of just 100.45 K, whereas the PINN's error is a staggering 465.00 K, indicating that our predictions are numerically far closer to the true values. More critically, the structural fidelity of the predictions highlights an even greater disparity. Our model's Pearson Correlation reaches a near-perfect 0.9822, confirming that it accurately reproduces the spatial structure of the temperature field. Conversely, the PINN's score of only 0.6333 for this metric confirms that its predicted field is not only inaccurate in value but also severely distorted in shape.

Table 7
Quantitative performance comparison of the MAM-PhyGNN and PINN models on a 100-step autoregressive prediction task. All metrics were computed on the unseen test set (Case 5).

Model variant	RMSE (K) ↓	MAE (K) ↓	Pearson Corr. ↑	Rel. Err. (%) ↓
Full model (Ours)	**100.45**	**40.91**	**0.9822**	**6.64**
PINN	465.00	400.79	0.6333	57.04

This dramatic performance disparity is fundamentally rooted in the different computational paradigms of the two approaches. A conventional PINN, by design, treats the physical domain as a collection of independent spatiotemporal points. It enforces physical laws by minimizing a PDE residual term within its loss function across this point cloud. This methodology, while elegant, has a critical limitation: it is topology-agnostic. It lacks an intrinsic mechanism to understand the local connectivity between points, which is the very foundation of heat conduction. Furthermore, the limitations of this topology-agnostic approach are severely exacerbated by the extreme sparsity of the computational mesh employed in our framework. To prioritize real-time inference, our graph utilizes only 2236 nodes. In such a sparse regime, relying solely on soft PDE residuals as a loss term provides insufficient regularization compared to traditional PINN applications, which typically employ hundreds of thousands of collocation points (e.g., 788 651 points [43]) to adequately bound the physical solution space. Consequently, for a dynamic process like a moving melt pool, where steep thermal gradients are constantly shifting, this architectural shortcoming becomes a major bottleneck. The model struggles to accurately resolve the local flow of heat, leading to the observed error accumulation and, ultimately, the divergence seen in long-term predictions.

In stark contrast, MAM-PhyGNN is architecturally designed to respect this local physics. By discretizing the domain into a graph, the model explicitly encodes the physical connectivity between neighboring regions. The message-passing scheme then directly simulates the local exchange of heat between nodes, perfectly aligning the model's computations with the underlying physical process. Furthermore, our dual-branch architecture provides a synergistic solution: the physics-hardcoded Laplace Block enforces the fundamental law of diffusion, providing a robust and stable foundation, while the data-driven GNN Block learns the complex, nonlinear dynamics that the simple diffusion model cannot capture.

In conclusion, this comparative experiment provides compelling evidence that our framework's topology-aware and physics-hardcoded design offers a fundamental advantage over the point-wise, soft-constraint approach of traditional PINNs. This makes MAM-PhyGNN a far more suitable and reliable tool for simulating transient thermal processes in metal additive manufacturing.

3.3.5. Computational performance

In addition to predictive accuracy and stability, a central motivation for developing a surrogate model is to dramatically reduce computational cost. To quantify this, we compared the inference time of our trained MAM-PhyGNN model against the runtime of the high-fidelity FLOW-3D simulation used to generate the ground-truth data.

Fig. 12 provides a direct comparison of the computational performance. The high-fidelity simulations were executed on a high-performance workstation equipped with dual Intel®Xeon®Platinum

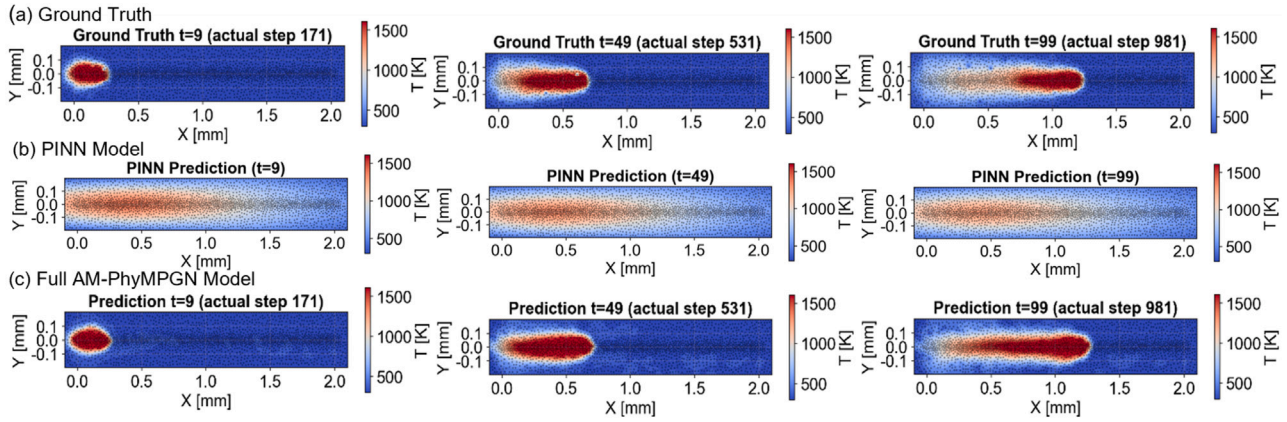


Fig. 11. Qualitative comparison of the MAM-PhyGNN and PINN models during a long-term autoregressive prediction. The figure displays temperature field snapshots at prediction steps 9, 49, and 99. (a) The Ground Truth field from the high-fidelity simulation. (b) The prediction from the conventional PINN, illustrating its catastrophic failure over time. (c) The prediction from our MAM-PhyGNN model, demonstrating its excellent long-term stability and physical fidelity.

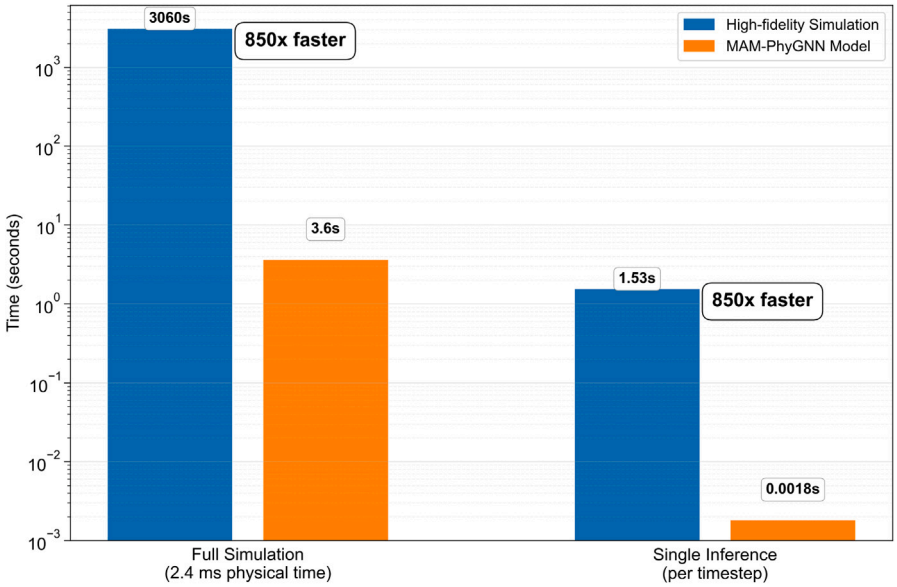


Fig. 12. Computational performance comparison. The y-axis (Time) is on a logarithmic scale to accommodate the significant performance difference. The model achieves an approximate 850-fold speedup in both scenarios.

8475B CPUs (totaling 192 logical processors). Even with this powerful hardware, the average time to complete a single 2.4 ms simulation case was 3060 s. In stark contrast, our trained MAM-PhyGNN model, running on a single NVIDIA A100-80GB GPU, performed the equivalent long-term prediction in just 3.6 s. This performance gap is even more pronounced at the single-timestep level. The high-fidelity solver required 1.53 s to compute a single step, whereas our model's forward pass (a single inference) took merely 0.0018 s. When evaluating the overall computational efficiency, the overhead of data preprocessing must also be considered. In the MAM-PhyGNN framework, the graph topology and the discrete Laplace matrix depend solely on the spatial geometry of the computational domain. Because they are fundamentally independent of transient thermal states and specific process parameters, such as laser power and scan speed, this structural data only needs to be generated once. For the 2236-node mesh utilized in this study, the initial graph construction required approximately 2.07 s, and computing the discrete Laplace matrix took 870.61 s. Once computed, these components are cached and completely reused for all subsequent long-term predictions and evaluations across different parameter combinations. By strictly decoupling this one-time offline geometric preprocessing from the online inference phase, the framework preserves its rapid evaluation capabilities without introducing redundant computational costs.

As highlighted in Fig. 12, this translates to a consistent computational speedup of approximately 850× across both full-simulation and per-timestep metrics. This order-of-magnitude acceleration, achieved while maintaining high physical fidelity as demonstrated in the preceding sections, confirms that MAM-PhyGNN successfully overcomes the fidelity-efficiency trade-off. This efficiency unlocks the potential for applications previously considered intractable for traditional solvers, such as large-scale parametric studies, real-time process control, and the development of high-fidelity digital twins.

4. Discussion

The results of this study clearly demonstrate that the proposed MAM-PhyGNN framework can predict the complex, transient thermal dynamics of MAM with a combination of speed, accuracy, and long-term stability that has been unattainable with prior surrogate modeling approaches. This exceptional performance, particularly the stability

in long-term autoregressive prediction, is a direct consequence of its unique dual-branch, physics-hardcoded architecture. This section will delve into the mechanisms behind the key experimental findings and elaborate on their significance for the field of AM simulation.

The most striking result from the ablation study was the catastrophic failure of the *GNN Block-only* model (Fig. 10(b)). This phenomenon highlights a fundamental issue: purely data-driven models, even when operating on graph structures, struggle to learn physical laws that can evolve stably over long periods from data alone without explicit physical constraints. In autoregressive prediction, the iterative feedback of microscopic inference errors forces the network to evaluate states that progressively deviate from its training distribution—a fundamental challenge termed exposure bias [44,45]. Because a pure GNN lacks an inherent thermodynamic smoothing mechanism—such as heat diffusion—to naturally dissipate these unnatural thermal gradients, its non-linear layers inadvertently non-linearly amplify these micro-errors rather than suppressing them. This causes the model to drift off the true physical manifold and eventually degenerate into a chaotic, physically meaningless state.

In contrast, our full model, through its *hardcoded* Laplace Block, embeds the core physics of linear heat diffusion as a non-learnable, deterministic operator. This Laplace Block acts as a powerful physical regularizer, providing a stable physical baseline for every prediction step. It effectively suppresses the exponential growth of errors, thereby ensuring stability over the nearly 1000-step rollout.

Concurrently, the performance of the *Laplace Block-only* model (Fig. 10(c)) reveals the limitations of a purely physics-based linear model. Although this variant remained stable, it failed to accurately capture the complex morphology of the moving melt pool, resulting in an unphysically bulbous and distorted shape at the leading edge. This directly proves that the linear diffusion equation is insufficient to describe the full spectrum of physics in the AM process. Complex effects such as the strong nonlinearities of temperature-dependent material properties (thermal conductivity, specific heat), the absorption and release of latent heat during phase change, and the radiative boundary conditions collectively constitute the *residual dynamics* of the physical model. Our GNN Block was designed precisely to learn this residual.

The results demonstrate that it successfully captured these nonlinear effects, providing the necessary corrections to the linear physics model and thus enabling a high-fidelity reproduction of the melt pool's shape and temperature distribution. Therefore, the success of MAM-PhyGNN is not rooted in physics or data alone, but in the synergistic interplay between the two, achieved through a clear division of labor.

The comparative experiment with the PINN (Fig. 11) further reveals the paradigmatic advantage of our approach. The failure of the PINN is rooted in its topology-agnostic nature. It treats the computational domain as a collection of independent points and *perceives* physics through a global loss function, but it ignores the fact that heat conduction is an inherently local process dependent on interactions between adjacent elements. For a dynamic system with a moving heat source and steep thermal gradients, this architectural flaw makes it difficult to effectively capture the local flow and transfer of heat, ultimately leading to the collapse of the prediction. Conversely, the graph structure of MAM-PhyGNN is inherently topology-aware. Its message-passing scheme perfectly mimics the exchange of heat between neighboring nodes, allowing the model to respect the locality of the physical process at an architectural level. This fundamental difference explains why our model remains stable and accurate on the same long-horizon forecasting task where the PINN fails.

Beyond accuracy and stability, the demonstrated ~850× computational speedup (from over 50 min to 3.6 s) has profound practical implications. This leap in efficiency makes applications previously intractable due to high computational costs now feasible. For instance, researchers can leverage the model to conduct large-scale parametric studies to rapidly generate process-structure maps for process optimization. Furthermore, its near-real-time prediction speed lays the groundwork for the development of digital twin systems capable of online monitoring, prediction, and even active control of the printing process—a critical step towards achieving intelligent and robust AM production.

5. Conclusion

In this work, we introduced and validated MAM-PhyGNN, a physics-hardcoded GNN for the fast and high-fidelity thermal simulation of the metal additive manufacturing process. Through its innovative dual-branch architecture, the model combines deterministic linear physics (heat diffusion) with data-driven nonlinear corrections (residual dynamics) on a graph structure that is aware of local physical interactions. Experimental results demonstrate that, compared to conventional PINNs and purely data-driven models, MAM-PhyGNN exhibits unparalleled stability and accuracy in long-horizon autoregressive predictions, while achieving a computational speedup of over two orders of magnitude. The core contribution of this study is the demonstration that embedding physical laws as *hardcoded* structural priors, rather than as *soft-constraint* loss terms, is key to building stable and reliable surrogate models for physical processes. The MAM-PhyGNN framework offers a potent solution to the long-standing fidelity-efficiency dilemma in AM and paves the way for the development of practical digital twin systems.

Despite its promising performance, the current study has specific limitations that define clear avenues for future research. Physically, the model simplifies complex melt pool phenomena by neglecting explicit powder consolidation and phase transition effects, focusing solely on transient thermal dynamics. Geometrically and operationally, the framework was validated in a 2D setting on a static mesh, focusing solely on single-track laser passes without incorporating real-world, complex scanning trajectory information. Consequently, at its present stage, MAM-PhyGNN functions primarily as an open-loop, forward-predictive surrogate model; thus, further development is required before it can actively guide the control of real-world printing processes. Future work will therefore focus on four critical areas: (1) extending the framework to full 3D simulations; (2) incorporating multi-physics coupling to capture melt pool convection and predict thermal stress; and (3) adapting the model to handle complex, multi-track scan strategies and dynamic graph structures to simulate the layer-by-layer fabrication of entire components; (4) evolving the current framework into a closed-loop dynamic control system. It is important to note that, at its present stage, MAM-PhyGNN functions primarily as a high-fidelity static surrogate model—serving essentially as a rapid forward predictive engine. While it successfully provides the requisite computational speed for real-time inference, its transition into active dynamic control, such as Model Predictive Control, will necessitate the incorporation of system identification methods. This future step is crucial to explicitly map process actuators to the dynamic state space, thereby bridging the gap between static thermal forecasting and active, closed-loop process adjustment. Successfully addressing these challenges will elevate the MAM-PhyGNN framework from a rapid simulation tool into a comprehensive predictive engine for optimizing the additive manufacturing process.

CRediT authorship contribution statement

Huaqing Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhibin Zhao:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Xingwu Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Xuefeng Chen:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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